

Skipping breakfast before and during early pregnancy and incidence of gestational diabetes mellitus: the Japan Environment and Children's Study

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ABSTRACT

Background: There is little evidence linking breakfast skipping to the incidence of gestational diabetes mellitus (GDM).

Objectives: We aimed to evaluate the prospective association of breakfast consumption with the incidence of GDM.

Methods: A total of 103,099 pregnancies were registered during study enrollment (January 2011 to March 2014), involving 97,454 pregnant women from 15 areas across Japan. Singleton pregnant women free of GDM, stroke, heart disease, cancer, and type 1 or type 2 diabetes at the study enrollment were eligible for analysis. We used a self-administered questionnaire to obtain data on demographic information, socioeconomic status, self-rated health status, disease history, lifestyle, and dietary habits of each woman at study enrollment. A semiquantitative FFQ was used for dietary assessment. We used logistic regression to obtain the OR of GDM in relation to breakfast consumption.

Results: Among 84,669 pregnant women for analysis, 1935 cases of GDM were documented. After adjustment for potential confounding factors including prepregnancy BMI, the multivariable-adjusted ORs of GDM for women who consumed breakfast 5–6 times/wk, 3–4 times/wk, and 0–2 times/wk were 1.09 (95% CI: 0.93, 1.27), 1.14 (95% CI: 0.96, 1.34), and 1.21 (95% CI: 1.05, 1.41), respectively, compared with daily breakfast eaters. The association appeared to be dose dependent (P -trend = 0.006) and was not significantly modified by prepregnancy BMI.

Conclusions: Breakfast consumption <3 times/wk before and during early pregnancy, compared with daily consumption, was associated with an increased odds of developing GDM. *Am J Clin Nutr* 2020;00:1–6.

Keywords: breakfast, pregnancy, gestational diabetes, cohort study, prevention

Introduction

The proportion of people who regularly consume breakfast has been decreasing in recent decades among adults in the

United States (1) and adolescents in many countries (2). Emerging evidence from observational studies indicates that skipping breakfast is associated with increased risk of many noncommunicable diseases, including obesity (3, 4), metabolic syndrome (3, 4), type 2 diabetes (5), stroke (6), and coronary heart disease (7).

Gestational diabetes mellitus (GDM) is a common complication during pregnancy that has various effects on both the mother and fetus. For instance, GDM significantly increases the risk of macrosomia and cesarean delivery (8, 9); in the long term, GDM is also associated with elevated risk of type 2 diabetes and cardiovascular diseases in the mother (10, 11) and a higher risk of obesity and insulin insensitivity in the offspring of mothers with GDM (12, 13). Clinical trials have suggested that dietary and exercise interventions could help prevent GDM (14), but little is known about the role of breakfast consumption in the development of GDM. Also, to the best of our knowledge, there are no guidelines on breakfast consumption for pregnant women or the general population. We herein aimed to examine the

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The findings and conclusions of this article are solely the responsibility of the authors and do not represent the official views of the above funder. Data described in the manuscript, code book, and analytic code will not be made available because of ethical restrictions and the legal framework of Japan. J-YD is the guarantor of this work and, as such, had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Supplemental Material is available from the "Supplementary data" link in the online posting of the article and from the same link in the online table of contents at <https://academic.oup.com/ajcn/>.

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Abbreviations used: GDM, gestational diabetes mellitus; HbA1c, glycated hemoglobin; OGTT, oral-glucose-tolerance test.

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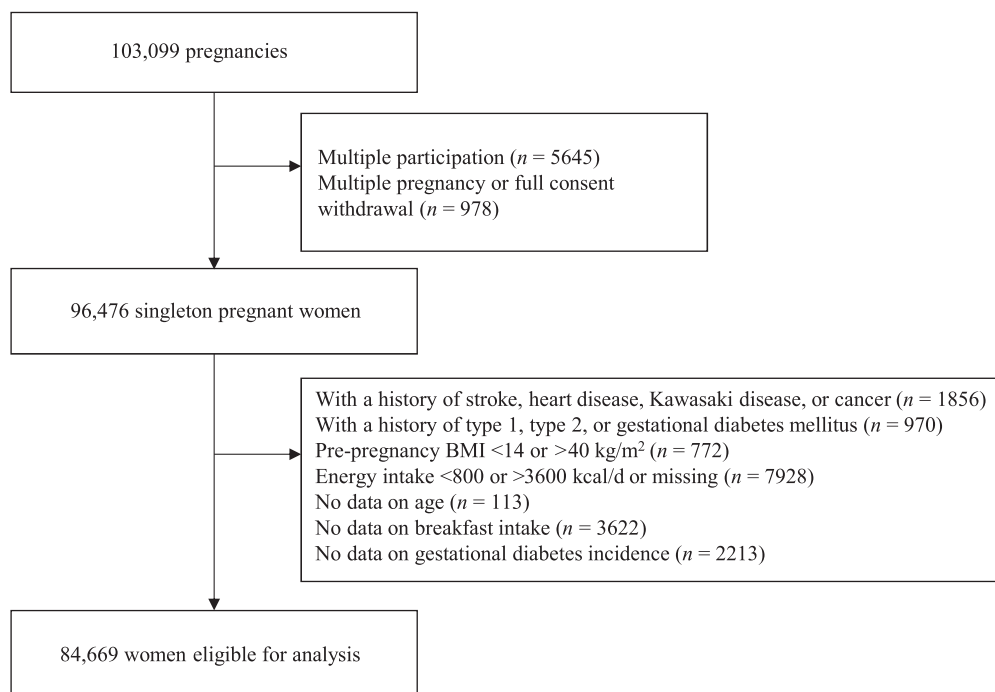


FIGURE 1 Flowchart for participant selection (there is overlap between the numbers of women excluded).

prospective association between breakfast skipping and incidence of GDM using a large Japanese national birth cohort.

Methods

The present analysis was based on the Japan Environment and Children's Study, a birth cohort launched by the Ministry of Environment with the primary objective to evaluate the effects of environmental factors on pregnancy and children's health. Details of the study design have been described elsewhere (15, 16). In brief, a total of 103,099 pregnancies were registered during study enrollment (January 2011 to March 2014), involving 97,454 pregnant women with a median gestational age of 12 wk (IQR = 5) from 15 areas across Japan. The characteristics of the women and their children, such as age at delivery, parity, gestational age at birth, sex ratio, and birth weight, showed high similarity to data from Japan's Vital Statistics Survey (16).

Women were eligible for the analysis if they had singleton pregnancies and were free of GDM, cardiovascular disease, Kawasaki disease, cancer, and type 1 or type 2 diabetes at study enrollment. Women were excluded if they had an extreme prepregnancy BMI (i.e., <14 or >40 kg/m²), reported implausible total energy intake (<800 or >3600 kcal/d), or provided no data on breakfast consumption, GDM incidence, or other critical variables. Eventually, a total of 84,669 women were included in the present analysis (Figure 1).

The original question about breakfast consumption was "How often did you have breakfast during the past 12 months?" The frequency included 6 categories: <1 time/mo, 1–3 times/mo, 1–2 times/wk, 3–4 times/wk, 5–6 times/wk, and every day. The first 3 categories were grouped as 0–2 times/wk because of a limited number of women in these categories.

Dietary assessment was based on a semiquantitative FFQ that has been shown as a useful instrument for dietary assessment among women in early pregnancy (17). Dietary patterns were identified from the FFQ by principal components analysis as previously reported (18). The prudent dietary pattern was characterized by increased consumption of soy foods, vegetables, mushrooms, seaweed, and seafood, and the Western dietary pattern was characterized by an increased consumption of beef, pork, processed meat, and high-fat milk. Other covariates including demographic information, socioeconomic status, self-rated health status, disease history (cardiovascular diseases, cancers, diabetes, mental diseases, obstetric/gynecological diseases, and so forth), and lifestyle were assessed based on self-reported questionnaires and medical records, which were detailed in previous reports (16, 19). The questionnaires were returned by mail at a median gestational age of 15 wk (IQR = 7).

The outcome of the current study was the first incidence of GDM. Medical records were used to identify cases. GDM was coded as "Yes" or "No" for each woman except for 2213 women, and women with missing data were excluded before the analysis (Figure 1). According to the 2011 guideline from the Japan Society of Obstetrics and Gynecology and the Japan Association of Obstetricians and Gynecologists, all pregnant women are recommended to have a screening test for GDM in early pregnancy (<14 gestational wk) and in midpregnancy (24–28 gestational wk) by random blood glucose or 50-g glucose challenge test (20). In our study, 77,387 women had data on blood glucose and 23,053 women had data on the 50-g glucose challenge test (data were missing in 4080 women). For those with a positive result in either early pregnancy or midpregnancy, a 75-g oral-glucose-tolerance test (OGTT) was then performed. GDM was diagnosed when ≥ 2 values during the 75-g OGTT were greater than the cutoff concentrations: fasting plasma

TABLE 1 Baseline characteristics of 84,669 singleton pregnant women by the frequency of breakfast consumption^{1,2}

	Every day	5–6 times/wk	3–4 times/wk	0–2 times/wk	P-trend
No. of women	58,737	8926	7333	9673	
Maternal age, y	31.5 (4.8)	29.5 (5.0)	28.9 (5.1)	28.6 (5.3)	<0.001
Gestational age at enrollment, ³ wk	12 (5)	12 (5)	12 (5)	12 (5)	0.68
Prepregnancy BMI, kg/m ²	21.0 (3.0)	21.3 (3.3)	21.5 (3.5)	21.5 (3.5)	<0.001
University or higher education, %	23.7	21.6	16.1	15.3	<0.001
Annual income <2 million yen, %	4.1	5.0	7.5	8.0	<0.001
Night-shift worker, %	6.3	10.4	10.7	12.2	<0.001
Married, %	96.7	94.2	92.4	89.5	<0.001
Nulliparous, %	36	53.1	50.9	57.3	<0.001
History of macrosomia babies, %	0.5	0.4	0.5	0.4	0.39
Current smoker, %	2.6	5.5	8.6	12.5	<0.001
Current drinker, %	8.9	9.3	9.6	9.2	<0.001
Depression, %	2.5	3.5	3.9	4.8	<0.001
Polycystic ovarian syndrome, %	2.3	2.3	2.2	1.7	<0.001
Physical activity, Met-h/d	3.8 (7.9)	3.8 (8.2)	4.3 (9.2)	4.3 (9.0)	<0.001
Total energy, kcal/d	1805 (536)	1738 (526)	1723 (549)	1641 (552)	<0.001
Prudent dietary pattern ⁴ score, % above the median	43.7	34.3	32.4	27.9	<0.001
Western dietary pattern ⁵ score, % above the median	38.9	42.0	42.6	42.7	<0.001

¹ Values are means (SDs) or percentage unless otherwise specified. Met-h/d, metabolic equivalent task hours/day.

² Generalized linear models were used for trend test.

³ Values are medians (IQR).

⁴ The prudent dietary pattern is characterized by increased consumption of soy foods, vegetables, mushrooms, seaweed, and seafood.

⁵ The Western dietary pattern is characterized by increased consumption of beef, pork, processed meat, and high-fat milk.

glucose ≥ 5.5 mmol/L (100 mg/dL), 1-h value ≥ 10.0 mmol/L (180 mg/dL), and 2-h value ≥ 8.3 mmol/L (150 mg/dL) (21).

The present analysis was based on the jecs-ag-20160424 dataset with data collected from study enrollment to 1 month after birth (released in June 2016 and revised in October 2016). Age-adjusted means and proportions of the characteristics of included pregnant women were calculated according to the frequency of breakfast consumption. Generalized linear models were used to perform trend tests across groups. Women consuming breakfast every day were treated as the reference group, and logistic regression was used to obtain the OR and 95% CI of GDM in comparison with the reference group. All ORs were adjusted for maternal age at baseline in a basic model (model 1). We also adjusted for study area (15 units), smoking status (never, past, and current smokers, or missing category), drinking status (never, past, current drinkers, or missing category), education level (middle school, high school, junior or specialized training college, university or higher, or missing category), occupation (15 categories, or missing category), household income (9 categories, or missing category), history of depression (yes or no), history of having infants with macrosomia (yes or no), history of polycystic ovarian syndrome (yes or no), marital status (married, divorced, widowed, other, or missing category), parity (0, 1, 2, ≥ 3 , or missing category), and physical activity (quintiles or missing category) in model 2. In model 3, we further adjusted for total energy intake and Western dietary pattern scores (all quintiles). In model 4, further adjustment was made for prepregnancy BMI (quintile). Missing values of covariates were treated as a separate category. Trend tests were performed by modeling breakfast consumption as a continuous variable and assigning the median value of each group (e.g., 1 for 0–2 times/wk and 7 for every day). We also conducted stratified analyses according to prepregnancy BMI (<25 or ≥ 25) to examine whether there was a modifying effect. Interaction tests were conducted by entering the

interaction terms in the logistic model. In addition, we performed sensitivity analyses by restricting the analysis to women who 1) kept the same frequency of breakfast consumption before and during pregnancy, 2) had data on GDM screening, 3) had a gestational age <14 wk at baseline, and 4) were not night-shift workers. SAS 9.4 (SAS Institute Inc) was used for analyses. All *P* values were 2-sided, and a *P* value <0.05 was considered statistically significant.

The study protocol was approved by the Institutional Review Board on Epidemiological Studies of the Ministry of the Environment. Written informed consent was obtained from each participant.

Results

Table 1 presents the baseline characteristics of women according to the frequency of breakfast consumption before and during early pregnancy. In this population, 69.4% of the women ate breakfast every day whereas 11.4% of the women ate breakfast 0–2 times/wk. There was no difference in gestational age at baseline across groups (median = 12 wk; IQR = 5 for all groups). Compared with daily breakfast consumers, women who skipped breakfast more frequently appeared to be younger and less educated, had a slightly higher BMI and lower household income, and were more likely to be current smokers but less likely to be current drinkers. These women were also more likely to be night-shift workers, unmarried, nulliparous, and have depression. As for diet, women who skipped breakfast had lower intakes of total energy, lower scores of prudent dietary pattern, but higher scores of Western dietary pattern.

We identified 1935 cases of GDM among 84,669 singleton pregnant women during follow-up. The median gestational age at GDM diagnosis was 28 wk (IQR = 9). **Table 2** shows the ORs (95% CIs) of GDM according to the frequency of

TABLE 2 ORs for gestational diabetes mellitus according to the frequency of breakfast consumption¹

	Every day	5–6 times/wk	3–4 times/wk	0–2 times/wk	<i>P</i> -trend
No. of women	58,737	8926	7333	9673	
No. of cases	1306	206	177	246	
Gestational age at diagnosis, ² wk	28 (9)	27 (8)	28 (7)	27 (9)	
Model 1	1	1.23 (1.06, 1.43)	1.36 (1.16, 1.59)	1.47 (1.28, 1.69)	<0.001
Model 2	1	1.15 (0.99, 1.34)	1.21 (1.03, 1.43)	1.27 (1.10, 1.48)	<0.001
Model 3	1	1.15 (0.98, 1.34)	1.20 (1.02, 1.42)	1.26 (1.09, 1.47)	<0.001
Model 4	1	1.09 (0.93, 1.27)	1.14 (0.96, 1.34)	1.21 (1.05, 1.41)	0.006

¹Logistic regression was used to obtain the ORs (95% CIs). Model 1: adjusted for maternal age at study baseline. Model 2: model 1 and further adjusted for study area (15 units), smoking status (never, past, and current smokers, or missing category), drinking status (never, past, current drinkers, or missing category), education level (middle school, high school, junior or specialized training college, university or higher, or missing category), occupation (15 categories or missing category), household income (9 categories, or missing category), history of depression (yes or no), history of having infants with macrosomia (yes or no), history of polycystic ovarian syndrome (yes or no), marital status (married, divorced, widowed, other, or missing category), parity (0, 1, 2, ≥3, or missing category), and physical activity (quintiles or missing category). Model 3: model 2 and further adjusted for total energy intakes and Western dietary pattern scores (all quintiles). Model 4: model 3 and further adjusted for prepregnancy BMI (quintiles).

²Values are medians (IQR).

breakfast consumption. Compared with women who consumed breakfast every day, the age-adjusted ORs (95% CIs) of GDM for women who consumed breakfast 5–6 times/wk, 3–4 times/wk, and 0–2 times/wk were 1.23 (1.06, 1.43), 1.36 (1.16, 1.59), and 1.47 (1.28, 1.69), respectively. Adjustment for demographic factors, lifestyle-related factors, and other risk factors (model 2), and further adjustment for Western dietary pattern scores and total energy intakes (model 3), yielded slightly attenuated but significant results. After further adjustment for prepregnancy BMI (model 4), the ORs (95% CIs) of GDM were 1.09 (0.93, 1.27), 1.14 (0.96, 1.34), and 1.21 (1.05, 1.41), respectively. Breakfast skipping was associated with an increased risk of GDM in a dose-dependent manner in all models (all *P*-trend ≤0.006).

Results from the stratified analysis indicated that prepregnancy BMI did not significantly modify the association between

breakfast consumption and risk of GDM. For example, the fully multivariable-adjusted ORs for women who consumed breakfast 0–2 times/wk were 1.21 (95% CI: 1.01, 1.45) and 1.16 (95% CI: 0.88, 1.52) among those with a prepregnancy BMI <25 and ≥25, respectively (*P*-interaction = 0.41). The sensitivity analyses showed similar results, although the association was borderline significant among women who kept the same frequency of breakfast consumption before and during pregnancy (Table 3).

Discussion

To the best of our knowledge, the present large prospective cohort study is the first study assessing breakfast consumption in relation to the incidence of GDM. We found that breakfast consumption <3 times/wk before and during early pregnancy was

TABLE 3 Sensitivity analyses for the association between breakfast consumption and gestational diabetes mellitus

	Every day	5–6 times/wk	3–4 times/wk	0–2 times/wk	<i>P</i> -trend
Keeping the same frequency of breakfast consumption before and during pregnancy					
No. of women	52,535	3676	2330	4454	
No. of cases	1175	72	56	108	
OR ¹	1.00	0.97 (0.76, 1.24)	1.22 (0.93, 1.62)	1.23 (0.99, 1.53)	0.04
Having data on gestational diabetes screening					
No. of women	55,970	8497	6930	9192	
No. of cases	1290	205	175	243	
OR ¹	1.00	1.10 (0.94, 1.28)	1.17 (0.99, 1.39)	1.24 (1.07, 1.44)	0.002
Gestational age <14 wk at baseline					
No. of women	39,620	5887	4910	6489	
No. of cases	927	146	125	173	
OR ¹	1.00	1.14 (0.94, 1.36)	1.18 (0.97, 1.43)	1.28 (1.07, 1.52)	0.003
Being not shift-workers					
No. of women	55,117	7982	6529	8467	
No. of cases	1232	185	146	221	
OR ¹	1.00	1.09 (0.93, 1.29)	1.07 (0.90, 1.29)	1.27 (1.09, 1.48)	0.004

¹Logistic regression was used to obtain the ORs (95% CIs), with adjustment for maternal age at study baseline, study area (15 units), smoking status (never, past, and current smokers, or missing category), drinking status (never, past, current drinkers, or missing category), education level (middle school, high school, junior or specialized training college, university or higher, or missing category), occupation (15 categories or missing category), household income (9 categories, or missing category), history of depression (yes or no), history of having infants with macrosomia (yes or no), history of polycystic ovarian syndrome (yes or no), marital status (married, divorced, widowed, other, or missing category), parity (0, 1, 2, ≥3, or missing category), physical activity (quintiles or missing category), total energy intakes, Western dietary pattern scores (all quintiles), and prepregnancy BMI (quintiles).

associated with an increased odds of developing GDM, which was independent of prepregnancy BMI.

The incidence rate of GDM in the present study (2.3%) was lower than that in other populations. The latest Japanese diagnostic criteria of diabetes mellitus, which is an adoption of the International Association of Diabetes and Pregnancy Study Groups (IADPSG) criteria, were released by the Japan Diabetes Society in July 2010 but not standardized for GDM diagnosis until August 2015 (22). Because the recruitment of our study ended in March 2014, the diagnoses of GDM were largely based on the previous criteria (21), which showed a low rate (2.4%) in an earlier report (23). Although GDM assessments were based on medical records, we were not able to verify the validation because of a lack of relevant data. In addition, using medical records as the only source for outcome assessment can lead to underestimation of number of cases and hence decreased statistical power.

Although little evidence to date has linked breakfast consumption to the risk of GDM, several studies have examined the association of eating breakfast with risk of lifestyle-related diseases. An increased risk of type 2 diabetes is consistently reported among breakfast skippers in previous studies, with a summarized RR of 1.33 (95% CI: 1.22, 1.46) in a meta-analysis of 6 prospective cohort studies (5). In a prospective cohort study among Swedish adolescents aged 16 y, higher incidences of central obesity, high fasting glucose, and metabolic syndrome were observed in those with poor breakfast habits, compared with regular breakfast consumers after 27 y of follow-up (4). A cross-sectional study of 4052 adults aged 40–54 y found that skipping breakfast was associated with a higher prevalence of atherosclerosis, independent of traditional cardiovascular risk factors (24). Also, breakfast skipping was shown to be associated with a ~20% increase in the risk of coronary heart disease in a US male cohort (7) and in the risk of stroke in a Japanese cohort (6).

Several possible mechanisms might underlie the association of breakfast skipping with risk of GDM. Short-term randomized controlled trials have found that breakfast skipping resulted in increased fat oxidation, higher postprandial insulin concentrations, and increased postprandial glycemic response, which could lead to low-grade inflammation and impaired glucose metabolism in the long term (25, 26). A cross-sectional study of 4116 UK primary-school children aged 9–10 y showed that compared with daily breakfast consumers, breakfast skippers had higher levels of insulin resistance and glycated hemoglobin (HbA1c) even after adjustment for adiposity markers (27). Numerous cross-sectional studies have shown a relation between breakfast skipping and an increased prevalence of overweight and obesity (28). Prospective cohort studies, in which a temporal relation can be established, found that skipping breakfast was a risk factor for weight gain and obesity, probably owing to increased hunger and overeating after a longer fasting time (3, 4, 29). Interestingly, there is also evidence that skipping breakfast can affect clock gene and clock-controlled gene expression and hence disrupt circadian rhythms in both healthy and diabetic individuals (25).

BMI has been suggested as a possible mediator in previous studies assessing the association between breakfast skipping and risk of type 2 diabetes. In the previously mentioned meta-analysis (5), when the results of 4 studies with adjustment for BMI were pooled, the overall RR for type 2 diabetes was attenuated from 1.33 (95% CI: 1.22, 1.46) to 1.22 (95% CI: 1.12, 1.34).

Similarly, our study showed a slightly attenuated result after further adjustment for prepregnancy BMI (OR from 1.26 to 1.21). These results were in line with prospective cohort studies showing that skipping breakfast was associated with weight gain and obesity (3, 4, 29). In contrast, a recent meta-analysis of 7 randomized controlled trials found no evidence supporting the notion that breakfast consumption helps with weight loss or that breakfast skipping leads to weight gain (30). In fact, participants who skipped breakfast experienced a small reduction in body weight in comparison with those who ate breakfast (30). However, the findings of that meta-analysis (30) should be treated with caution owing to limitations of the individual trials, including lack of blinding, limited number of participants ($n < 50$ in most trials), and short intervention duration (mean = 7 wk, range = 2–16 wk).

The strengths of the present study included a large sample size and a prospective cohort design. However, several limitations should also be addressed. First, we could not rule out the influence of residual confounding, which is inevitable in observational studies. As shown in Table 1, breakfast skippers were more likely to smoke, have lower socioeconomic status, and adopt a Western dietary pattern, which could be a threat to the internal validity of our study. Despite careful control for known risk factors and potential confounding factors, it is uncertain to what extent other unmeasured factors related to breakfast skipping could affect the observed association; therefore, residual confounding remains an alternative explanation for the observed association. Second, we were unable to examine the relation of breakfast quality with risk of GDM because of the absence of data on breakfast contents. There is evidence that higher breakfast quality based on the Alternate Healthy Eating Index 2010, which measures conformance to dietary guidelines (31), is associated with a healthier cardiovascular profile and lower concentrations of HbA1c in men (32). Third, women might change their eating habits to healthier ones during the period of pregnancy. A second survey showed that among the 9673 women who ate breakfast 0–2 times/wk before and during early pregnancy, 5219 women increased their consumption frequency during mid-to-late pregnancy. Our sensitivity analysis restricted to women keeping the same frequency of breakfast consumption before and during pregnancy yielded a slightly weakened association, which was probably due to the decreased number of cases. Fourth, our study was prone to measurement bias because the questionnaire asked about breakfast consumption during the past 12 mo. Of note, such bias was speculated to occur equally in all groups, and the association was therefore probably underestimated.

In conclusion, we found that breakfast consumption <3 times/wk before and during early pregnancy, compared with daily consumption, was associated with an increased odds of developing GDM. Further studies among other populations are warranted, and studies assessing breakfast quality in relation to risk of GDM could further our understanding of the importance of breakfast consumption.

Members of the Japan Environment and Children's Study are listed in the online **Supplemental Material**.

The authors' contributions were as follows—J-YD: designed the study, analyzed the data, and wrote the manuscript; Takashi Kimura, SI, HI: conducted the technique review and reviewed and edited the manuscript; MC, YK, Tadashi Kimura, KU: reviewed the manuscript; Takashi Kimura, SI, KU,

HI: helped to collect the data; and all authors: read and approved the final manuscript. The authors report no conflicts of interest.

References

- Haines PS, Guilkey DK, Popkin BM. Trends in breakfast consumption of US adults between 1965 and 1991. *J Am Diet Assoc* 1996;96(5): 464–70.
- Lazzeri G, Ahluwalia N, Niclasen B, Pammolli A, Vereecken C, Rasmussen M, Pedersen TP, Kelly C. Trends from 2002 to 2010 in daily breakfast consumption and its socio-demographic correlates in adolescents across 31 countries participating in the HBSC study. *PLoS One* 2016;11(3):e0151052.
- Odegaard AO, Jacobs DR Jr, Steffen LM, Van Horn L, Ludwig DS, Pereira MA. Breakfast frequency and development of metabolic risk. *Diabetes Care* 2013;36(10):3100–6.
- Wennberg M, Gustafsson PE, Wennberg P, Hammarstrom A. Poor breakfast habits in adolescence predict the metabolic syndrome in adulthood. *Public Health Nutr* 2015;18(1):122–9.
- Ballon A, Neuenschwander M, Schlesinger S. Breakfast skipping is associated with increased risk of type 2 diabetes among adults: a systematic review and meta-analysis of prospective cohort studies. *J Nutr* 2019;149(1):106–13.
- Kubota Y, Iso H, Sawada N, Tsugane S. Association of breakfast intake with incident stroke and coronary heart disease: the Japan Public Health Center-based study. *Stroke* 2016;47(2):477–81.
- Cahill LE, Chiuve SE, Mekary RA, Jensen MK, Flint AJ, Hu FB, Rimm EB. Prospective study of breakfast eating and incident coronary heart disease in a cohort of male US health professionals. *Circulation* 2013;128(4):337–43.
- Kc K, Shakya S, Zhang H. Gestational diabetes mellitus and macrosomia: a literature review. *Ann Nutr Metab* 2015;66(Suppl 2): 14–20.
- He XJ, Qin FY, Hu CL, Zhu M, Tian CQ, Li L. Is gestational diabetes mellitus an independent risk factor for macrosomia: a meta-analysis? *Arch Gynecol Obstet* 2015;291(4):729–35.
- Bellamy L, Casas JP, Hingorani AD, Williams D. Type 2 diabetes mellitus after gestational diabetes: a systematic review and meta-analysis. *Lancet* 2009;373(9677):1773–9.
- Li J, Song C, Li C, Liu P, Sun Z, Yang X. Increased risk of cardiovascular disease in women with prior gestational diabetes: a systematic review and meta-analysis. *Diabetes Res Clin Pract* 2018;140:324–38.
- Boerschmann H, Pfluger M, Henneberger L, Ziegler AG, Hummel S. Prevalence and predictors of overweight and insulin resistance in offspring of mothers with gestational diabetes mellitus. *Diabetes Care* 2010;33(8):1845–9.
- Maftai O, Whitrow MJ, Davies MJ, Giles LC, Owens JA, Moore VM. Maternal body size prior to pregnancy, gestational diabetes and weight gain: associations with insulin resistance in children at 9–10 years. *Diabet Med* 2015;32(2):174–80.
- Shepherd E, Gomersall JC, Tieu J, Han S, Crowther CA, Middleton P. Combined diet and exercise interventions for preventing gestational diabetes mellitus. *Cochrane Database Syst Rev* 2017;11:CD010443.
- Kawamoto T, Nitta H, Murata K, Toda E, Tsukamoto N, Hasegawa M, Yamagata Z, Kayama F, Kishi R, Ohya Y, et al. Rationale and study design of the Japan Environment and Children's Study (JECS). *BMC Public Health* 2014;14:25.
- Michikawa T, Nitta H, Nakayama SF, Yamazaki S, Isobe T, Tamura K, Suda E, Ono M, Yonemoto J, Iwai-Shimada M, et al. Baseline profile of participants in the Japan Environment and Children's Study (JECS). *J Epidemiol* 2018;28(2):99–104.
- Ogawa K, Jwa SC, Kobayashi M, Morisaki N, Sago H, Fujiwara T. Validation of a food frequency questionnaire for Japanese pregnant women with and without nausea and vomiting in early pregnancy. *J Epidemiol* 2017;27(5):201–8.
- Nanri A, Shimazu T, Ishihara J, Takachi R, Mizoue T, Inoue M, Tsugane S; Group JFVS. Reproducibility and validity of dietary patterns assessed by a food frequency questionnaire used in the 5-year follow-up survey of the Japan Public Health Center-Based Prospective Study. *J Epidemiol* 2012;22(3):205–15.
- Michikawa T, Nitta H, Nakayama SF, Ono M, Yonemoto J, Tamura K, Suda E, Ito H, Takeuchi A, Kawamoto T, et al. The Japan Environment and Children's Study (JECS): a preliminary report on selected characteristics of approximately 10 000 pregnant women recruited during the first year of the study. *J Epidemiol* 2015;25(6): 452–8.
- Minakami H, Hiramatsu Y, Koresawa M, Fujii T, Hamada H, Iitsuka Y, Ikeda T, Ishikawa H, Ishimoto H, Itoh H, et al. Guidelines for obstetrical practice in Japan: Japan Society of Obstetrics and Gynecology (JSOG) and Japan Association of Obstetricians and Gynecologists (JAOG) 2011 edition. *J Obstet Gynaecol Res* 2011;37(9):1174–97.
- Kuzuya T, Nakagawa S, Satoh J, Kanazawa Y, Iwamoto Y, Kobayashi M, Nanjo K, Sasaki A, Seino Y, Ito C, et al. Report of the Committee on the Classification and Diagnostic Criteria of Diabetes Mellitus. *Diabetes Res Clin Pract* 2002;55(1):65–85.
- Hiramatsu Y, Haneda M, Yasuhi I, Namba M, Uchigata Y, Ueki K, Atsumi Y, Watada H. [The joint committee with the Japan Society of Diabetes and Pregnancy and the Japan Diabetes Society: an abnormal glucose metabolism during pregnancy and the standardization of its diagnostic criteria]. *Journal of the Japan Diabetes Society* 2015;58(10):801–3, Japanese.
- Morikawa M, Yamada T, Yamada T, Akaishi R, Nishida R, Cho K, Minakami H. Change in the number of patients after the adoption of IADPSG criteria for hyperglycemia during pregnancy in Japanese women. *Diabetes Res Clin Pract* 2010;90(3):339–42.
- Uzhova I, Fuster V, Fernandez-Ortiz A, Ordovas JM, Sanz J, Fernandez-Friera L, Lopez-Melgar B, Mendiguren JM, Ibanez B, Bueno H, et al. The importance of breakfast in atherosclerosis disease: insights from the PESA study. *J Am Coll Cardiol* 2017;70(15):1833–42.
- Jakubowicz D, Wainstein J, Landau Z, Raz I, Ahren B, Chapnik N, Ganz T, Menaged M, Barnea M, Bar-Dayana Y, et al. Influences of breakfast on clock gene expression and postprandial glycemia in healthy individuals and individuals with diabetes: a randomized clinical trial. *Diabetes Care* 2017;40(11):1573–9.
- Nas A, Mirza N, Hagele F, Kahlhofer J, Keller J, Rising R, Kufer TA, Bosy-Westphal A. Impact of breakfast skipping compared with dinner skipping on regulation of energy balance and metabolic risk. *Am J Clin Nutr* 2017;105(6):1351–61.
- Donin AS, Nightingale CM, Owen CG, Rudnicka AR, Perkin MR, Jebb SA, Stephen AM, Sattar N, Cook DG, Whincup PH. Regular breakfast consumption and type 2 diabetes risk markers in 9- to 10-year-old children in the Child Heart and Health Study in England (CHASE): a cross-sectional analysis. *PLoS Med* 2014;11(9): e1001703.
- Horikawa C, Kodama S, Yachi Y, Heianza Y, Hirasawa R, Ibe Y, Saito K, Shimano H, Yamada N, Sone H. Skipping breakfast and prevalence of overweight and obesity in Asian and Pacific regions: a meta-analysis. *Prev Med* 2011;53(4-5):260–7.
- van der Heijden AA, Hu FB, Rimm EB, van Dam RM. A prospective study of breakfast consumption and weight gain among U.S. men. *Obesity (Silver Spring)* 2007;15(10):2463–9.
- Sievert K, Hussain SM, Page MJ, Wang Y, Hughes HJ, Malek M, Cicuttini FM. Effect of breakfast on weight and energy intake: systematic review and meta-analysis of randomised controlled trials. *BMJ* 2019;364:142.
- Chiuve SE, Fung TT, Rimm EB, Hu FB, McCullough ML, Wang M, Stampfer MJ, Willett WC. Alternative dietary indices both strongly predict risk of chronic disease. *J Nutr* 2012;142(6):1009–18.
- Iqbal K, Schwingshackl L, Gottschald M, Knuppel S, Stelmach-Mardas M, Aleksandrova K, Boeing H. Breakfast quality and cardiometabolic risk profiles in an upper middle-aged German population. *Eur J Clin Nutr* 2017;71(11):1312–20.